Multimedia Signal Processing - 2nd module

Prof. Marco Tagliasacchi

September, 15 2011

1 Question - 10 pts

Consider a gray-scale image with 3 grey levels $L = 0, 1, 2$. The following statistics can be estimated collecting data from an image collection:

$$p(0) = \Pr(l = 0) = 0.25$$
$$p(1) = \Pr(l = 1) = 0.35$$
$$p(2) = \Pr(l = 2) = 0.40$$

- Compute the entropy of the source.
- Design a Huffman code using block sizes $N = 1, 2$.
- In each case, compute the average code length L_N.

Inter-pixel correlation is described by the following statistics

$$P = \begin{bmatrix} 0.10 & 0.10 & 0.05 \\ 0.05 & 0.25 & 0.05 \\ 0.10 & 0.00 & 0.30 \end{bmatrix} \tag{1}$$

where p_{ij} denotes the probability $\Pr(l_1 = i, l_2 = j)$, given two consecutive pixels l_1 and l_2 in raster scan order.

- Show that pixel values of consecutive pixels are statistically dependent
- Compute the joint entropy
- Show how to compute the conditional probabilities that need to be used in an arithmetic coder. Explicitly compute $\Pr(l_1 = 0|l_2 = 0)$.

Solution

The entropy of the source is

$$H(X) = -0.25 \log_2 0.25 - 0.35 \log_2 0.35 - 0.40 \log_2 0.40 = 1.5589 \text{ bps} \tag{2}$$

An optimal Huffman code for $N = 1$ is

$$C(0) = 10$$
$$C(1) = 11$$
$$C(2) = 0$$

and the corresponding average code length $L_1 = 2 \cdot 0.25 + 2 \cdot 0.35 + 0.40 = 1.6$ bps.

In order to design an optimal Huffman code for $N = 2$, we need to form super-symbols and compute the joint probabilities. Assuming a discrete memoryless source, we obtain

$$\Pr(l_1 = 0, l_2 = 0) = 0.0625$$
$$\Pr(l_1 = 0, l_2 = 1) = 0.0875$$
$$\Pr(l_1 = 0, l_2 = 2) = 0.1000$$
$$\Pr(l_1 = 1, l_2 = 0) = 0.0875$$
$$\Pr(l_1 = 1, l_2 = 1) = 0.1225$$
$$\Pr(l_1 = 1, l_2 = 2) = 0.1400$$
$$\Pr(l_1 = 2, l_2 = 0) = 0.1000$$
$$\Pr(l_1 = 2, l_2 = 1) = 0.1400$$
$$\Pr(l_1 = 2, l_2 = 2) = 0.1600$$

An optimal Huffman code for $N = 2$ is

$$C(00) = 1100$$
$$C(01) = 1101$$
$$C(02) = 001$$
$$C(10) = 000$$
$$C(11) = 011$$
$$C(12) = 100$$
$$C(20) = 010$$
$$C(21) = 101$$
$$C(22) = 111$$

The corresponding average code length $L_2 = 3.15/2 = 1.5750$ bps. Note that the Huffman code is not unique. However, all Huffman codes will have the same optimal code length.
Pixel values of consecutive pixels are not statistically independent. Indeed, it suffices to show that $\exists i, j$ such that $\Pr(l_1 = i, l_2 = j) \neq \Pr(l_1 = i)\Pr(l_1 = j)$. For example, consider $i = j = 0$.
The joint entropy is $H(X, Y) = 2.666 \leq 2 \cdot 1.5589 = 3.1178$.
The conditional probabilities can be computed given the marginal and the joint probabilities. That is

$$\Pr(l_1 = i | l_2 = j) = \frac{\Pr(l_1 = i, l_2 = j)}{\Pr(l_2 = j)} \tag{3}$$

For example, $\Pr(l_1 = 0 | l_2 = 0) = 0.10/0.25 = 0.4$

2 Question - 5 pts

Illustrate the block diagram of a lossy video coding architecture based on transform coding (both encoder and decoder).

Solution

See slide set `MMSP_3.2_Video coding`.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a discrete memoryless source with finite alphabet, and $D(R)$ the corresponding rate-distortion function, σ_x^2 its variance and $H(X)$ its entropy.

   - [T] [F] $H(X) < \infty$
     TRUE. $H(X) \leq \log_2 M < \infty$, where M is the number of symbols.
   - [T] [F] $D(0) = \sigma_x^2$
     TRUE. When $R = 0$ bits are allocated, the decoder reconstructs the source as a constant signal, equal to its expected value
   - [T] [F] $R(0) \geq H(X)$
     TRUE. When a discrete memoryless source is lossless coded ($D = 0$), the average code length is lower bounded by the entropy of the source.

2. Consider a perceptual audio coder

   - [T] [F] The analysis filter bank is the same used by the perceptual model
     FALSE. The analysis filterbank might be different from the one used by the perceptual model. The latter is typically designed to match the time-to-frequency mapping characteristic of the human ear. The former, instead, needs to be satisfy other requirements (e.g. perfect reconstruction, fast implementation, etc.).
   - [T] [F] Noise maskers have higher SMR than tonal maskers.
     FALSE. The intensity of masking is more pronounced for noise maskers.
   - [T] [F] All critical bands have the same bandwidth.
     FALSE. Critical bands are conventionally selected in such a way that their bandwidth increases when the center frequency increases, in such a way to match the higher frequency resolution at lower frequencies.
   - [T] [F] Noise and tonal maskers are lossless coded
     FALSE. Maskers above the masking threshold are quantized (lossy coded) using a number of bits such that the resulting quantization noise does not exceed the SMR.

3. Consider a lossy image coding system

   - [T] [F] The computational complexity of encoding is significantly higher than decoding.
     FALSE. Most operations are performed symmetrically on both encoder and decoder. More sophisticated rate allocation strategies might slightly increase the complexity of the encoder.
   - [T] [F] Coding efficiency increases when the block size increases.
     TRUE. A larger block size enables to capture inter-pixel correlations at further distance.
   - [T] [F] The DCT transform achieves perfect reconstruction (up to arithmetic precision)
     TRUE. The DCT transform is an orthonormal transform. Thus it is invertible.

4 Question - 12 pts

Local image features are routinely used in several computer vision applications, including augmented reality (e.g Google Goggles). Their success is due to ability to summarize the input image in a way that is invariant to several image transformations (scale, rotation, luminance, viewpoint etc...).

- Load the file `feature_set.mat`. It contains a collection of 128-dimensional SIFT local features extracted from a test image. Each dimension of each feature vector is quantized with 8 bits.
- Assume the feature set as a memoryless source that produces symbols from 0:255. What is the minimum number of bits to lossless encode the feature set?

Recent studies have shown that SIFT feature vectors are robust to heavy quantization, i.e. the accuracy of object recognition does not decrease when local features are lossy coded.

- Design a uniform quantizer with 2 reconstruction levels and compute the resulting MSE.
- Design an optimal quantizer with 2 reconstruction levels using the Lloyd-Max algorithm. Plot the resulting MSE for each iteration of the algorithm.

Solution

```
% Load the file feature_set. It contains a collection of 128-dimensional
% SIFT local features extracted from a test image. Each dimension of each feature
% vector is quantized with 8 bits.
clear all;
close all;
clc;

load feature_set;

% 1. Assume the feature set as a memoryless source that produces symbols from 0:2^8 -1.
% What is the minimum number of bits/dimension to encode the feature set without any loss?

sig = sig(:);
h = hist(sig,0:2^8-1);
p = h/sum(h);
p = p(p~=0);
H = -sum(p.*log2(p));

disp(['We can encode the feature set using: ' num2str(H) ' bits/dimension']);


% Recent studies have shown that SIFT feature vectors are robust to heavy quantization,
% i.e. the accuracy of object recognition does not decrease when lossy coding local features.

%2. Design a uniform quantizer with 2 reconstruction levels and compute the resulting MSE.

%simple uniform quantizer.
M = 2;
threshold_un = 255/2
rec_level_un = ceil([0 threshold_un] + threshold_un/2)

sig_q_uniform = NaN*ones(1,length(sig));
for i=1:length(sig)
    if(sig(i)<threshold_un)
        sig_q_uniform(i) = rec_level_un(1);
    else
        sig_q_uniform(i) = rec_level_un(2);
    end
end

MSE_uniform = var(double(sig')-double(sig_q_uniform))


%3. Design an optimal quantizer with 2 reconstruction levels using the Lloyd-Max algorithm.
%Plot the resulting MSE for each iteration of the algorithm.

threshold_opt = 255/2;
NUM_ITER = 20;
sig_q_optimal = NaN*ones(1,length(sig));
for iter=1:NUM_ITER

    %compute centroid of the pdf (reconstruction levels) restricted to the decision interval
    c1 = sum(sig(sig<threshold_opt))/length(sig(sig<threshold_opt));
    c2 = sum(sig(sig>threshold_opt))/length(sig(sig>threshold_opt));
    rec_level_opt = [c1 c2];

    %compute the new threshold
    threshold_opt = (c2 + c1)/2;

    for i=1:length(sig)
        if(sig(i)<threshold_opt)
            sig_q_optimal(i) = rec_level_opt(1);
        else
            sig_q_optimal(i) = rec_level_opt(2);
        end
    end

    %compute the MSE for the current iteration
    MSE_optimal(iter) = var(double(sig')-double(sig_q_optimal));
```

```
end
threshold_opt
rec_level_opt
plot(MSE_optimal);
ylabel('MSE');
xlabel('Iteration');
```